

A Holder-Datum Counterexample to Strict Hilbert Contractivity

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Source and Open Question

The source paper is Max Fathi, Samuel Sra, and Melanie Weber, *Computing Brascamp-Lieb Constants through the lens of Thompson Geometry*, arXiv:2208.05013. In Section 6, after introducing the normalized Picard map

$$\tilde{G}(X) = \frac{G(X)}{\|G(X)\|},$$

for an arbitrary matrix norm, the authors ask whether $\tilde{G}$ is strictly contractive with respect to Hilbert's projective metric. The relevant source passage is reproduced below.

One can show that $\tilde{G}$ is non-expansive and asymptotically regular with respect to δ_H and that it has a (not necessary unique) fixed point in the compact set $\tilde{D} := \{X : \frac{X}{\|X\|}\}$. Asymptotic convergence of $\tilde{G}$ is then guaranteed by a Hilbert geometry version of Theorem 4. Such a result can be obtained by specializing the more general Banach space version of (Baillon et al., 1978, Theorem 1.2) to Hilbert geometry.

We can show that the Picard iteration defined by the map $\tilde{G}$ converges to the same fixed-point as that of the map G : Let X^* denote a fixed point of $\tilde{G}$. Then $G(X^*) = \lambda X^*$ for some $\lambda > 0$. Eq. 6.1 together with Corollary 34 imply that $\lambda = 1$, which gives the claim.

It remains an open question to show, whether $\tilde{G}$ is strictly contractive with respect to the Hilbert projective norm. This would allow to adapt our non-asymptotic convergence analysis to the map $\tilde{G}$ as well.

Proof Intuition

Hilbert's projective metric ignores positive scalar multiples. Therefore the normalization in $\tilde{G} = G/\|G\|$ cannot create projective contraction if the underlying map G is already the identity on a feasible datum. The Holder datum with repeated identity maps gives exactly this situation.

Definitions

For positive definite matrices $X, Y \in \mathbb{P}_d$, Hilbert's projective metric is

$$\delta_H(X, Y) = \log(M(X/Y)M(Y/X)), \quad M(X/Y) = \inf\{\lambda > 0 : X \preceq \lambda Y\}.$$

The source Picard map is

$$G(X) = \left(\sum_{j=1}^m w_j A_j (A_j^* X A_j)^{-1} A_j^* \right)^{-1}.$$

Strict contractivity means

$$\delta_H(\tilde{G}(X), \tilde{G}(Y)) < \delta_H(X, Y)$$

for all projectively distinct positive definite matrices X, Y in the domain.

Counterexample

Theorem 1. *The unqualified assertion that $\tilde{G}$ is strictly contractive in Hilbert's projective metric is false for feasible Brascamp-Lieb data.*

Proof. Fix any dimension $d \geq 2$ and consider the Holder Brascamp-Lieb datum

$$m = 2, \quad A_1 = A_2 = I_d, \quad w_1 = w_2 = \frac{1}{2}.$$

This is feasible: it is the standard Holder datum from the source paper, with $\sum_j w_j = 1$. Equivalently, Tao's feasibility conditions give

$$d = \sum_{j=1}^2 w_j d$$

and, for every subspace $H \subseteq \mathbb{R}^d$,

$$\dim H = \sum_{j=1}^2 w_j \dim(I_d H).$$

For this datum, the Picard map is exactly the identity:

$$\begin{aligned} G(X) &= \left(\frac{1}{2} I_d (I_d^* X I_d)^{-1} I_d^* + \frac{1}{2} I_d (I_d^* X I_d)^{-1} I_d^* \right)^{-1} \\ &= \left(\frac{1}{2} X^{-1} + \frac{1}{2} X^{-1} \right)^{-1} = X. \end{aligned}$$

Hence

$$\tilde{G}(X) = \frac{X}{\|X\|}$$

for every positive definite X and every matrix norm used in the normalization.

Hilbert's projective metric is invariant under positive scalar rescaling: for all $\alpha, \beta > 0$,

$$\delta_H(\alpha X, \beta Y) = \delta_H(X, Y).$$

Indeed $M(\alpha X / \beta Y) = (\alpha / \beta) M(X / Y)$ and $M(\beta Y / \alpha X) = (\beta / \alpha) M(Y / X)$, so the two scalar factors cancel. Therefore

$$\delta_H(\tilde{G}(X), \tilde{G}(Y)) = \delta_H(X, Y)$$

for every pair of projectively distinct positive definite matrices X, Y .

For an explicit pair in dimension 2, take

$$X = \text{diag}(2, 1), \quad Y = I_2.$$

Then $M(X/Y) = 2$ and $M(Y/X) = 1$, so $\delta_H(X, Y) = \log 2 > 0$, and equality of distances proves that $\tilde{G}$ is not strictly contractive. $\square$

Remark 1. *This counterexample is intentionally narrow. The Holder datum above is feasible but not simple: every nonzero proper subspace H is critical, since*

$$\dim H = \sum_{j=1}^2 w_j \dim(I_d H).$$

Thus the example refutes the literal general feasible-datum formulation of the Section 6 question, but it does not settle a possible intended version restricted to simple Brascamp-Lieb data.

Verification Notes

The script `code/verify_holder_identity.py` checks the displayed 2×2 witness using the diagonal formula

$$\delta_H(\text{diag}(x_1, x_2), \text{diag}(y_1, y_2)) = \log \frac{\max_i x_i/y_i}{\min_i x_i/y_i}.$$

It prints equality of the pre- and post-iteration distances, both equal to $\log 2$.

Novelty and Limitations

Local run indexes were searched for arXiv:2208.05013, the paper title, Brascamp-Lieb, Thompson geometry, Hilbert projective metric, and strict contractivity. A bounded web search on 2026-07-03 for the exact open-question phrases, arXiv id, and close variants found no separate answer source. Because the example is elementary, novelty confidence is only moderate.

Human review should check whether the intended source question silently assumes simple data. Under that stronger reading, this packet should be read as a literal-scope caveat, not as a solution of the simple-data strict contractivity problem.

References

- [1] Max Fathi, Samuel Sra, and Melanie Weber. *Computing Brascamp-Lieb Constants through the lens of Thompson Geometry*. arXiv:2208.05013.